PSD301M - Project guidelines

1. **Project Objective**

This project helps you apply your Python knowledge step by step — from the basics, through data structures, NumPy, and Pandas, all the way to collecting real data from an API. By the end, you will demonstrate how Python can turn raw API data into meaningful insight.

2. **Project ideas:**

   Your team(4-5 students) will select whatever the domain knowledge that you would like to discover. Some subject recommendations are listed below:

   - Education: Student scores, attendance, dropout prediction.
   - Finance: Stock market or cryptocurrency data.
   - Weather: Historical weather dataset, temperature trends.
   - Sports: Player/team performance stats
   - …..

3. **Phase 1: Python Basics (Week 1-2)**

   Your team will apply knowledge regarding variables, loops, conditionals, and functions to proceed at least 4 requirements (the number of requirements depends on your team's members) based on your selected domain.

   For example, this is a requirement concerning the education domain that could apply Python function to revolve:

   1. Write a function that calculates if a student passes or fails based on their score.

   2….

4. **Phase 2: Data Structures (Week 3-4)**

   Use Python's core data structures to organize and manipulate data. Based on the type of data structures, your team should propose suitable requirements, at least one requirement for one data structure.

   Lists →for storing multiple records.
   Dictionaries → for structured data (like JSON objects from an API).
   Sets/Tuples → for unique/immutable data.

   Example: Store student info in a list of dictionaries.

5. **Phase 3: Data Collection from an API and analyze data with Numpy & Pandas (Week 5-8)**

   Your team will find a resource to collect data based on domain knowledge that you have chosen. Then:

   - Collect data from the optional resource:
     - Option A (API): Use a RESTful API (e.g., Twitter, OpenWeather, COVID-19, or financial APIs). Fetch data using requests and parse JSON responses.
     - Option B (Web Scraping):Scrape a website (e.g., Wikipedia tables, news headlines) using BeautifulSoup/Scrapy.
   - Organize into Pandas DataFrame.
   - Analyze with NumPy & Pandas.
     - Compute statistics (mean, median, correlation).
     - Generate 3 visualizations (Matplotlib/Seaborn):
     - Line/bar plot
     - Histogram/box plot
     - Scatter/heatmap
     - Explain insights in Jupyter Notebook markdown.
     - Skills Tested:

       ✓ Pandas (describe, corr)

       ✓ Matplotlib/Seaborn (plot, hist)

       ✓ Data storytelling
   - Save final results into CSV/Excel.

6. **Deliverables (Week 9-10)**

   6.1. Python Notebook/Script including:

   ✓ Basic Python exercises.

   ✓ Data structure examples.

   ✓ NumPy operations.

   ✓ Pandas data analysis.

- ✓ API data collection + analysis.

6.2. Processed dataset (CSV/Excel) exported from Pandas.

6.3. Mini report (1–2 pages) with:

- ✓ Which API you used and why.
- ✓ Steps of analysis.
- ✓ Key insights discovered.

7. Grading Rubrics

Criteria	Weight
Python basics (functions, loops, conditionals)	10%
Data structures (lists, dicts, sets/tuples)	15%
NumPy analysis (arrays, stats, calculations)	15%
Pandas usage (DataFrame, filtering, grouping)	20%
API data collection & integration	25%
Code readability & final report	15%